



Test plan and report

TPR-group4_carotidStent_performance_Nr1

	Name	Date
Author	Natalya Ryzhuk, Yevheniia Ivanova, Carla Zumkehr	9.06.2026
Supervisor		

1. Objective

To assess and compare the engineering performance of two stents generations by

- A: Determining the force needed to bend the stent by 25°
- B: Determining the starting point of ovalization (80% of the initial diameter)

in two different conditions: fully-expanded vs. implanted in a hydrogel model

2. Materials

2.1 Test material

Description of the material to be tested (implant, model...) with indication of number of samples, batch...

- Nitinol stents manufactured by admedes, laser cut:
 - 8mm diameter
 - 50mm length
 - Generation 1: rigid (one sample)
 - Generation 2: flexible (one sample)

Control:

No control material is used in this study. Acceptance criteria are defined based on literature.

2.2 Model material

Description of the (anatomical, standardized...) model used to perform the test.

The development of the model was realized by segmenting a renal artery in 3D-slicer and designing a model for the hydrogel.

- Hydrogel model
 - One sample
 - Length:
 - Diameter:



Remarks:

- The pre-designed mold was produced on the 28th of May and 3D-printed in the polymer lab in the Institute of Biomedical Engineering
- The hydrogel model will be manufactured one week before testing by molding it into the pre-designed mold. Further information will be provided once the manufacturing process is finished

Justification:

- Hydrogel as model material reproduces the vessel mechanics and provides high transparency for investigating stent properties

Justification: *Size and/or materials of the model must be justified and classified as best- or worst-case model, if applicable.*

2.3 Material for procedure and equipment

Description of disposable material (tubes, connectors, syringes...) and equipment (sensors, pumps, microscopes...)

- Load cell with amplifier board (HX711-01, joy-it)
- Linear actuator with stepper motor (DM542T(V4.0), stepperonline)
- Digital microscope (provided by BMT)
- 3D-printed mount for fixing the stent
- 3D-printed loading tip
- 3D-printed stopper for the linear actuator

2.4 Control material

Description of any well-known material used as a golden standard comparison for test material.

No control material

3. Methods

3.1 Graphical description

Schematic description of test setup

3.2 Procedure steps

Every test will be performed three times.

Procedure A: investigating the bending force:

1. Fix linear actuator in vertical position
2. Fix the load cell on the sled of the linear actuator
3. Mount the loading tip on the load cell at center of the stent



4. Fix the stent on the mount
5. Connect linear actuator to the computer
6. Start the stepper motor program
7. Move the sled of the linear actuator until first contact with the stent
8. Move down in 0.5mm steps
9. Record the needed force for every step
10. Stop at bending of the 2mm
11. Document the results

Procedure B: investigating the starting point of ovalization

1. Prepare microscope
2. Fix linear actuator in vertical position
3. Fix the load cell on the sled of the linear actuator
4. Mount the loading step on the load cell
5. Fix the stent on the mount
6. Connect linear actuator to the computer
7. Start the stepper motor program
8. Notate the starting diameter of the stent d_0
9. Move the sled of the linear actuator until first contact with the stent
10. Move down in 0.5-1mm steps
11. Take microscope image of cross-section d_x after every step x
12. Calculate the ratio of the resulting diameter via $\frac{d_0}{d_x}$
13. Starting point of ovalization: ratio of diameter equals 0.8
14. Record the force needed for each step

3.3 Measurements

List of all parameters measured with indication of time point. Sketch can be included.

Procedure A: investigating the bending force:

Number of measurement	Force [N]

Procedure B: investigating the starting point of ovalization

Step x	Diameter d_x	Ratio of diameter $\frac{d_0}{d_x}$	Force [N]

3.4 Observations

List of all features observed with indication of time point. Sketch can be included.



Parameter	Time point

4. Test conditions

- Room temperature between 16°C and 20°C
- Operator skill level based on the knowledge gained in lecture and practice

5. Acceptance criteria

Description of expected results

A: The bending force of the stent must be within 196 mN and 402 mN

B: The force measured in part A must not lead to an ovalization of less than 80% of the diameter d_0

Justification:

Part A: Values suggested in “Floris Vos et al, carotid stent mobility with regard to head movements: n vitro analysis, 2004”

Part B: The values are based on considerations of an acceptable ovalization level and taken from part A.

6. Limitations

- No consideration of arterial pulsation
- No bloodstream
- Stent is not in contact with blood
- No special geometries or consideration of atherosclerosis
- Measurements are only taken on one part of the stent
- Lack of comparability between the hydrogel and a human artery

7. Results

Description of results using tables, charts and pictures.

8. Deviation

Description of deviation from the original test plan, e.g. sample number, test conditions...



9. Protocol

Entry of all notes taken during the test